

# Tom Benest

*Ph.D. in Planetary Science*

📍 Hefei, China    ✉ tom.benest@ustc.edu.cn

## PROFILE

My research activity focus on the origins of the solar system and its giant planets. I work on the evolution of volatile species and the fate of the complex organic molecules in the protosolar nebula.

To do so, I modelled the evolution of the disk and the transport of the particles inside. The aim of my thesis is to assess the formation conditions of the planets and their moons in the Solar System, and therefore to prepare the scientific return of the ESA/JUICE and NASA/Europa-Clipper missions.

## EDUCATION

|                                  |                                                                                                                                                                                                                                                                                       |
|----------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2022 – 2025<br>Marseille, France | <b>Ph.D. in Physics and Materials Sciences, specializing in Astrophysics and Cosmology, LAM, Aix-Marseille Université</b><br>Study of the composition of the protosolar nebula and its role in the formation of the Giant planets and the Galilean moons. Supervisor: Olivier Mousis. |
| 2020 – 2022<br>Bordeaux, France  | <b>Master Fundamental Physic and Application (equivalent to MSC Degree), Université de Bordeaux</b><br>Specialisation: Nuclei, Plasma, and Universe.                                                                                                                                  |
| 2019 – 2020<br>Poitiers, France  | <b>Bachelor in Physics, Université de Poitiers</b>                                                                                                                                                                                                                                    |

## PROFESSIONAL EXPERIENCE

|                                        |                                                                                                                                                                                                                                                                                                            |
|----------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 05/2026 – Present<br>Hefei, China      | <b>Postdoctoral position, USTC</b><br>Study of the formation of giant planets and icy moons, and of the fate of organics in the primordial nebula.                                                                                                                                                         |
| 2022 – 2025<br>Marseille, France       | <b>Planetary Science Ph.D., LAM, Aix-Marseille Université</b><br>Study of the formation of the Solar System and its giant system. Modeling and simulation of protoplanetary disks to find constraints on the formation of planets such as Jupiter or Uranus, and their moons. Supervisor : Olivier Mousis. |
| 03/2022 – 06/2022<br>Marseille, France | <b>2nd year Master internship, Laboratoire d'Astrophysique de Marseille</b><br>Computer modeling and simulation of the transport and the irradiation of particles in a protoplanetary disk. Supervisor : Olivier Mousis.                                                                                   |
| 02/2022<br>La Mongie, France           | <b>Exoplanets direct imaging and transit photometry, Pic du Midi observatory</b><br>Use of the Pic du Midi observatory's T50 telescope to capture transits of exoplanets and image objects in the Messier catalogue, as part of a 2nd year Master project.                                                 |
| 04/2021 – 06/2021<br>Bordeaux, France  | <b>1st year Master internship, Laboratoire d'Astrophysique de Bordeaux</b><br>Exploitation of the code FARGO 3D with Python in order to simulate the AB-Aurigae system's protoplanetary disk. Supervisor: Emmanuel Di Folco.                                                                               |

## CONFERENCE, WORKSHOP & MEETING ATTENDANCE

|                                        |                                                                                                                 |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------|
| 09/2025<br>Helsinki, Finland           | <b>EPSC 2025</b><br>Europlanet Science Congress conference. Oral presentation and poster.                       |
| 04/2025<br>Vienna, Austria             | <b>EGU 2025</b><br>European Geosciences Union conference. Poster presentation.                                  |
| 01/2024 – 11/2024<br>Marseille, France | <b>Journal Club</b><br>Host of the team's weekly Journal Club, a meeting to discuss scientific news and papers. |
| 09/2024<br>Berlin, Germany             | <b>EPSC 2024</b><br>Europlanet Science Congress conference. Oral presentation and poster.                       |
| 06/2024<br>Marseille, France           | <b>SF2A 2024</b><br>Société Française d'Astronomie et d'Astrophysique conference. Oral Presentation.            |
| 04/2024<br>Vienna, Austria             | <b>EGU 2024</b><br>European Geosciences Union conference. Poster presentation.                                  |

|                               |                                                                                                                                               |
|-------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------|
| 04/2024<br>Göttingen, Germany | <b>JUICE-SWI science meeting</b><br>Scientific meeting of the JUICE-SWI team. Oral presentation.                                              |
| 12/2023<br>Toulouse, France   | <b>FACOM science meeting</b><br>Meeting with the consortium FACOM (Fate of the volatile COMpounds at the galilean Moons). Oral presentations. |
| 10/2023<br>Marseille, France  | <b>Workshop on the Origins and Habitability of the Galilean Moons</b><br>Organized by the Institut Origines. LOC member. Poster presentation. |
| 08/2023<br>Marseille, France  | <b>International Planetary Probe Workshop (IPPW) 2023</b><br>LOC member. Poster presentation.                                                 |
| 05/2023<br>La Palma, Spain    | <b>BEACON 2023</b><br>Biennial European Astrobiology Conference. Poster presentation.                                                         |
| 01/2023<br>Marseille, France  | <b>FACOM science meeting</b><br>Meeting with the consortium FACOM (Fate of the volatile COMpounds at the galilean Moons). Oral presentation.  |

## PUBLICATIONS

|         |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|---------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2026    | <b>Tracing Uranus and Neptune Formation with Disequilibrium Species,</b><br><i>Astronomy and Astrophysics, in prep.</i><br>Tom Briand, Vincent Hue, Olivier Mousis, Thibault Cavalié, <b>Tom Benest Couzinou</b> ,<br>Antoine Schneeberger, Mark D. Hofstadter                                                                                                                                                                                                                                                                                                                                                                                                   |
| 2026    | <b>The Deuterium-to-Hydrogen ratio in Uranus as a marker of its formation conditions,</b><br><i>Astronomy &amp; Astrophysics, in prep.</i><br><b>Tom Benest Couzinou</b> and Olivier Mousis.                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 2026    | <b>The Seven Mysteries of Callisto: A Science Case for a Next-Generation Orbiter and Lander Mission,</b> <i>Science China, submitted</i><br>Michel Blanc, Francis Nimmo, Jingyan Cai, Zhonghua Yao, Jihua Hao, Daniel Arnold, Jean-Pierre Barriot, Yufeng Lin, Federico Tosi, Shane Carberry Mogan, Thomas Le Liboux, Chi Wang, Zhi-Yang Liu, <b>Tom Benest Couzinou</b> , Bilal Benmahi, William Desprats, Marie Devinat, Vincent Hue, Taichi Kawamura, Wenya Li, Jian Liu, Philippe Lognonné, William Rapin, Lorenz Roth, Tianran Sun, Yuxian Wang, Jonas Rabia, Sebastien Rodriguez, Yiran Wang, Nicolas Andre, Christian Mazelle, Boding Ouyang, Jianguo Yan |
| 02/2026 | <b>Formation and Survival of Complex Organic Molecules in the Jovian Circumplanetary Disk,</b> <i>PSJ, Volume 7, Number 2, 41</i><br>Olivier Mousis, Clément Petetin, <b>Tom Benest Couzinou</b> , Antoine Schneeberger, and Yannis Bennacer                                                                                                                                                                                                                                                                                                                                                                                                                     |
| 01/2026 | <b>Delivery of Complex Organic Molecules to the system of Jupiter,</b><br><i>MNRAS, Volume 545, Issue , id.staf2074,</i><br><b>Tom Benest Couzinou</b> , Alizée Amsler Moulanier, and Olivier Mousis.                                                                                                                                                                                                                                                                                                                                                                                                                                                            |
| 04/2025 | <b>Primordial Origin of Methane on Eris and Makemake Supported by D/H Ratios,</b><br><i>The Astrophysical Journal Letters, Volume 983, Issue 1, Number 1</i><br>Olivier Mousis, Aaron Werlen, <b>Tom Benest Couzinou</b> , Antoine Schneeberger.                                                                                                                                                                                                                                                                                                                                                                                                                 |
| 12/2024 | <b>Journey of complex organic molecules: formation and transport in protoplanetary disks,</b> <i>Astronomy &amp; Astrophysics, Volume 692, A10</i><br><b>Tom Benest Couzinou</b> , Olivier Mousis, Grégoire Danger, Antoine Schneeberger, Artyom Aguichine, and Alexis Bouquet.                                                                                                                                                                                                                                                                                                                                                                                  |
| 08/2024 | <b>Insights on the Formation Conditions of Uranus and Neptune from Their Deep Elemental Compositions,</b> <i>Planetary Science Journal, Volume 5, Number 8, 173</i><br>Olivier Mousis, Antoine Schneeberger, Thibault Cavalié, Kathleen E. Mandt, Artyom Aguichine, Jonathan I. Lunine, <b>Tom Benest Couzinou</b> , Vincent Hue, Raphaël Moreno.                                                                                                                                                                                                                                                                                                                |
| 06/2024 | <b>Recipes for Forming a Carbon-Rich Giant Planet,</b><br><i>Space Science Reviews, Volume 220, Issue 4, number 44</i><br>Olivier Mousis, Thibault Cavalié, Jonathan I. Lunine, Kathleen E. Mandt, Ricardo Hueso, Artyom Aguichine, Antoine Schneeberger, <b>Tom Benest Couzinou</b> , David H. Atkinson, Vincent Hue, Mark Hofstadter, Udomlerd Srisuchinwong.                                                                                                                                                                                                                                                                                                  |

## TEACHING EXPERIENCE

---

|                                                             |                                                                                                                                                                                                                                                                                              |
|-------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 10/2023 – 10/2024<br>Observatoire de Haute-Provence, France | <b>SIMO internship mentoring</b><br>Introductory training in observation methods at Haute Provence Observatory.<br>16h of teaching in October to 1st and 2nd year masters students in fundamental and atmospheric physics on the use of the T80 telescope and astronomical image processing. |
| 09/2024 – 10/2024<br>Marseille, France                      | <b>2 months internship supervision</b><br>Co-supervision of a 2nd year Master student on the Evolution of the D/H ratio in the Protosolar Nebula.                                                                                                                                            |
| 02/2024 – 07/2024<br>Marseille, France                      | <b>5 months internship supervision</b><br>Co-supervision of a 2nd year Master student for 5 months, focused on the Survival of the Organic Matter in the Circumjovian Disk.                                                                                                                  |
| 01/2024 – 04/2024<br>Marseille, France                      | <b>Science watch mentoring</b><br>Co-supervision of a master student for a Science Watch project, entitled Delivery of Organic Matter to the Icy Moons' Primordial Hydrosphere.                                                                                                              |

## SKILLS

---

### Programming

*Python, Fortran, LATEX*

### Expertise in Planetary Science

*Protoplanetary disk, Planetary formation*

## LANGUAGES

---

### French

*Native language*

### English

*Fluent*

### Spanish

*Beginner*